COSC3106: Theory of Computing

Lec#8: Converting RE to NFA

School of Computer Science and Technology

Algoma UNIVERSITY

This lecture

- This lecture, will learn…
  - How to convert REs to regular languages

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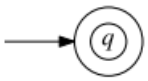
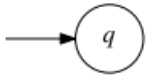
- **Definition: Any language can be described by a RE is a regular language.**
  - **RE is just a different tool that describe a regular language**

- Converting RE to an NFA:
  - Let Σ be a non-empty alphabet.

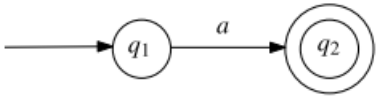
1. ε is a regular expression $\overset{we\ need}{\Longleftrightarrow}$ NFA that only accepts an empty string:



2. ∅ is a regular expression $\overset{we\ need}{\Longleftrightarrow}$ NFA that accepts ∅:



3. $a \in \Sigma, a$ is a regular expression $\overset{we\ need}{\Longleftrightarrow}$ NFA that accepts a:



4. $R_1 \cup R_2$ is a RE (apply union construction of NFA)
5. $R_1 R_2$ is a RE (apply concatenation construction of NFA)
6. R^* is a RE ((apply star construction of NFA)

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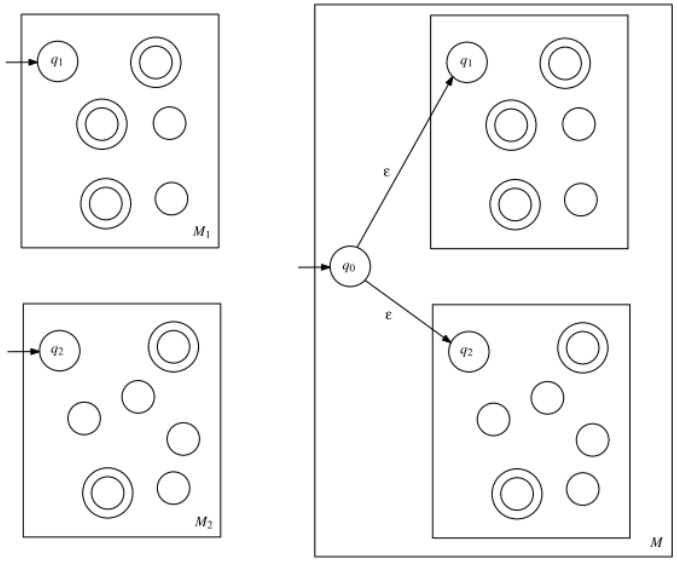
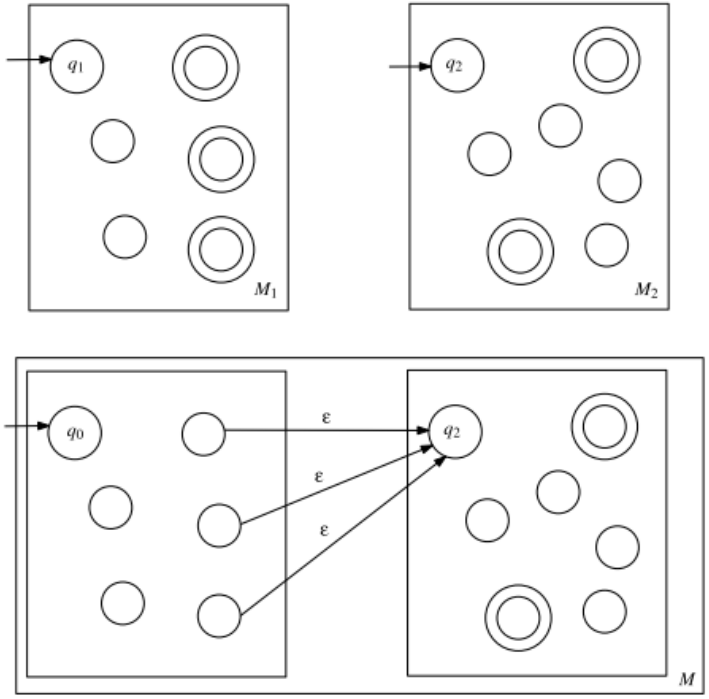
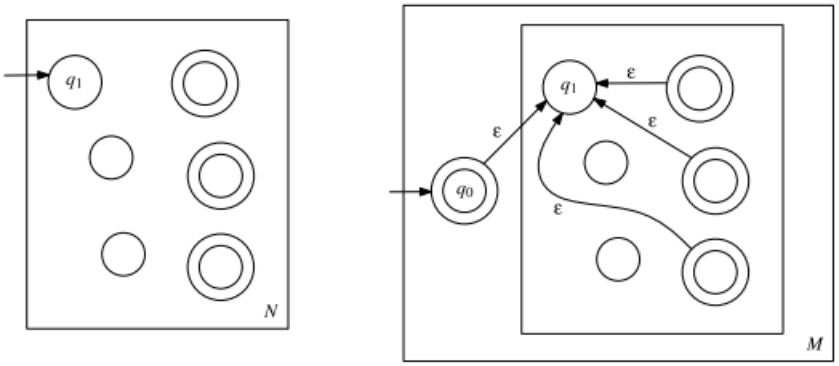

q_1
M_1
q_2
M_2
q_0
ε
ε
q_1
q_2
M


q_1
M_1
q_2
M_2
q_0
ε
ε
ε
q_2
M

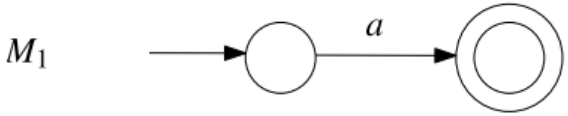

q_1
N
q_0
ε
q_1
ε
ε
ε
M

- Converting RE to an NFA:
  - Ex: If the the alphabet is $\{a, b\}$, convert the following RE to NFA:

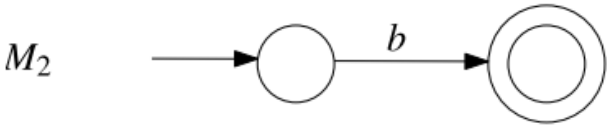
$$(ab \cup a)^*$$

We will prove that this regular expression describes a regular language, by constructing an NFA that accepts the language described by this regular expression.

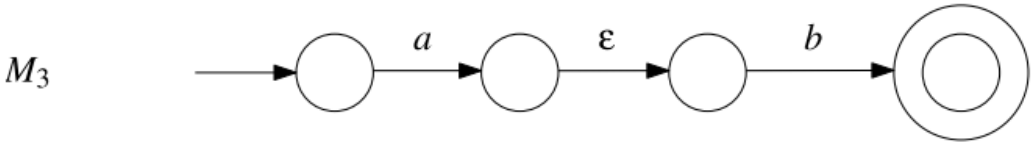
- First, we construct an NFA M_1 that accepts the language described by the regular expression a:



- Next, we construct an NFA M_2 that accepts the language described by the regular expression b:



- Next, we apply the concatenation of M_1 and M_2 . This gives an NFA M_3 that accepts the language described by the regular expression ab:



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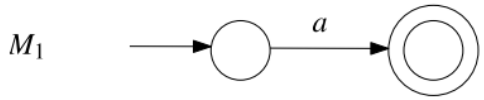
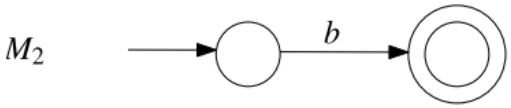
- **Converting RE to an NFA:**
  - **Ex: If the the alphabet is $\{a, b\}$, convert the following RE to NFA:**

$$(ab \cup a)^*$$

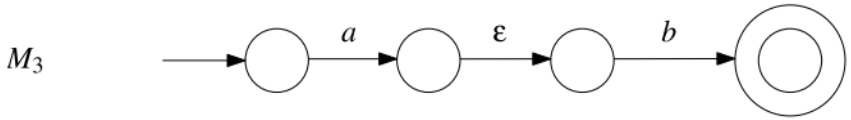
- **First, we construct an NFA M_1 that accepts the language described by the regular expression a:**



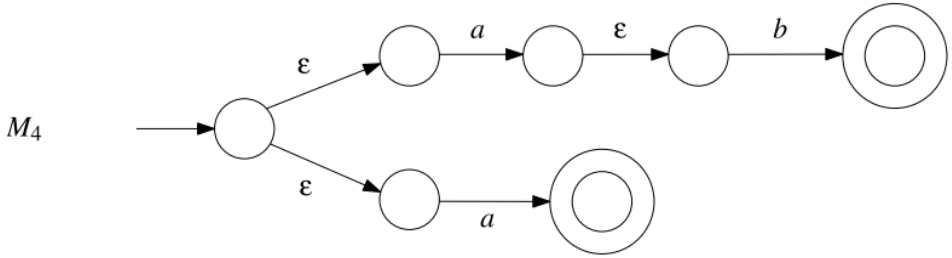
- **Next, we construct an NFA M_2 that accepts the language described by the regular expression b:**



- **Next, we apply the concatenation of M_1 and M_2. This gives an NFA M_3 that accepts the language described by the regular expression ab:**



- **Then, apply the construction to M_3 and M_1. This gives an NFA M_4 that accepts the language described by the regular expression $ab \cup a$:**



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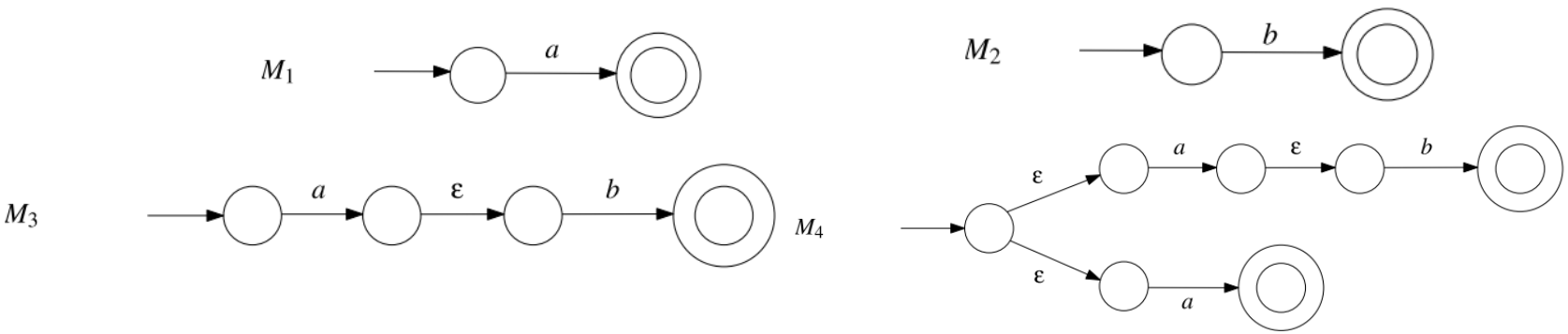
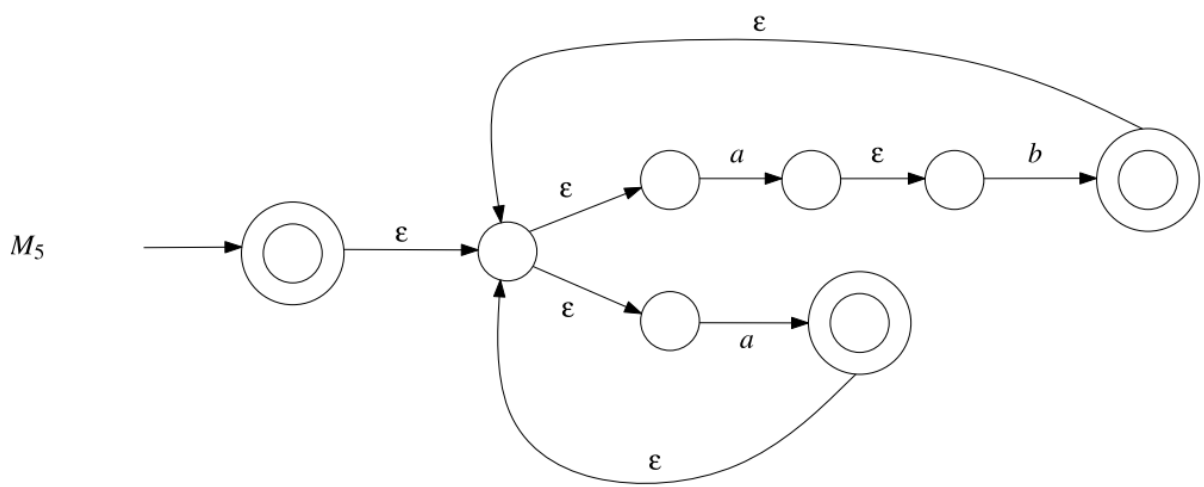
Converting RE to an NFA:

- Ex: If the the alphabet is $\{a, b\}$, convert the following RE to NFA:

$$(ab \cup a)^*$$



- Finally, we apply the construction of star to M_4 . This gives an NFA M_5 that accepts the language described by the regular expression $(ab \cup a)^*$



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